Physics 2110 Solutions

HW # 01 1.9, 12, 41, 85, 87, 89, 95, 97

1.9 (a) $550 \text{ miles/gallon} = (550 \text{ miles/gallon})\left(\frac{1 \text{ km}}{0.6214 \text{ mile}}\right)\left(\frac{1 \text{ gallon}}{3.788 \text{ L}}\right)$

$= 23.4 \text{ km/L}$

(b) The volume of gas required is $\frac{1500 \text{ km}}{23.4 \text{ km/L}} = 64.1 \text{ L}$, $\frac{64.1 \text{ L}}{45 \text{ L/tank}} = 1.4 \text{ tank}$

1.12 $(365.24 \text{ days}/1 \text{ yr})\left(\frac{24 \text{ h}}{1 \text{ day}}\right)\left(\frac{3600 \text{ s}}{1 \text{ h}}\right) = 3.15567\cdots \times 10^7 \text{ s}$

$\pi \times 10^7 \text{ s} = 3.14159\cdots \times 10^7 \text{ s}$

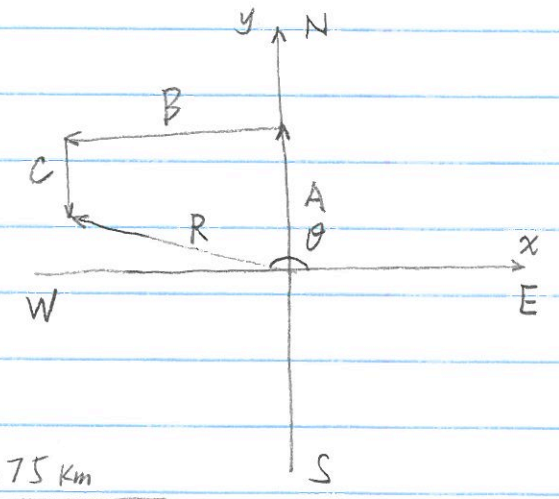
The approximate expression is accurate to two significant figures.

The percent error is 0.45%

1.41 $A_x = 0$, $A_y = +3.25 \text{ km}$

$B_x = -4.75 \text{ km}$, $B_y = 0$

$C_x = 0$, $C_y = -1.50 \text{ km}$



$R_x = A_x + B_x + C_x$

$R_x = 0 - 4.75 \text{ km} + 0 = -4.75 \text{ km}$

$R_y = A_y + B_y + C_y$

$R_y = 3.25 \text{ km} + 0 - 1.50 \text{ km} = 1.75 \text{ km}$

$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(-4.75 \text{ km})^2 + (1.75 \text{ km})^2} = 5.06 \text{ km}$

$\tan\theta = \frac{R_y}{R_x} = -0.3684$

$\theta = 159.8°$

The resultant displacement is 20.2° N of W

1.85 (a) With $A_z = B_z = 0$, Eq. (1.21) becomes

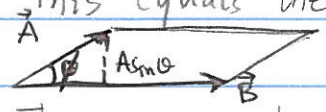
$$\vec{A}\cdot\vec{B} = A_x B_x + A_y B_y = (A\cos\theta_A)(B\cos\theta_B) + (A\sin\theta_A)(B\sin\theta_B)$$
$$= AB(\cos\theta_A\cos\theta_B + \sin\theta_A\sin\theta_B) = AB\cos(\theta_A - \theta_B) = AB\cos\phi$$

(b) With $A_z = B_z = 0$, $\vec{C} = C_z\vec{k}$ and $C = |C_z|$. From Eq (1.27),

$$C = |C_z| = |A_x B_y - A_y B_x| = |(A\cos\theta_A)(B\sin\theta_B) - (A\sin\theta_A)(B\cos\theta_B)|$$
$$= AB\,|\cos\theta_A\sin\theta_B - \sin\theta_A\cos\theta_B| = AB\,|\sin(\theta_B - \theta_A)| = AB\sin\phi$$

1.87 (a) The length of the base is B and the height of the parallelogram is $A\sin\phi$, so the area is $\underline{AB\sin\phi}$.

This equals the magnitude of the cross product.



(b) The cross product $\vec{A}\times\vec{B}$ is perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$, so the angle is 90°

1.89 (a) $A = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(-2.00)^2 + (3.00)^2 + (4.00)^2} = 5.38$

$B = \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(3.00)^2 + (1.00)^2 + (-3.00)^2} = 4.36$

(b) $\vec{A} - \vec{B} = (-2.00\vec{i} + 3.00\vec{j} + 4.00\vec{k}) - (3.00\vec{i} + 1.00\vec{j} - 3.00\vec{k})$

$\vec{A} - \vec{B} = (-2.00 - 3.00)\vec{i} + (3.00 - 1.00)\vec{j} + (4.00 + 3.00)\vec{k} = -5.00\vec{i} + 2.00\vec{j} + 7.00\vec{k}$

(c) Let $\vec{C} = \vec{A} - \vec{B}$ so $C_x = -5.00$, $C_y = 2.00$, $C_z = 7.00$

$C = \sqrt{C_x^2 + C_y^2 + C_z^2} = 8.83$

$\vec{B} - \vec{A} = -(\vec{A} - \vec{B})$, they have the same magnitude but opposite directions.

1.95 $\vec{C}$ is perpendicular to $\vec{A}$, so $\vec{A}\cdot\vec{C}=0$

$A_x C_x + A_y C_y = 0 \qquad 5.0\,C_x - 6.5\,C_y = 0$

$\vec{B}\cdot\vec{C} = 15.0$, so $-3.5\,C_x + 7.0\,C_y = 15.0$

Solving these two equations gives $C_x = 8.0$, $C_y = 6.1$

1.97 (a) $\vec{A}\cdot(\vec{B}\times\vec{C}) = A_x(\vec{B}\times\vec{C})_x + A_y(\vec{B}\times\vec{C})_y + A_z(\vec{B}\times\vec{C})_z$

$= A_x(B_y C_z - B_z C_y) + A_y(B_z C_x - B_x C_z) + A_z(B_x C_y - B_y C_x)$

$(\vec{A}\times\vec{B})\cdot\vec{C} = (\vec{A}\times\vec{B})_x C_x + (\vec{A}\times\vec{B})_y C_y + (\vec{A}\times\vec{B})_z C_z$

$= (A_y B_z - A_z B_y) C_x + (A_z B_x - A_x B_z) C_y + (A_x B_y - A_y B_x) C_z$

Compare $\vec{A}\cdot(\vec{B}\times\vec{C})$ and $(\vec{A}\times\vec{B})\cdot\vec{C}$, they contain the same terms, so $\vec{A}\cdot(\vec{B}\times\vec{C}) = (\vec{A}\times\vec{B})\cdot\vec{C}$

(b) $A = 5.00$, $\theta_A = 26.0^\circ$; $B = 4.00$, $\theta_B = 63.0^\circ$

$|\vec{A}\times\vec{B}| = AB\sin\phi \qquad \phi = \theta_B - \theta_A = 63.0^\circ - 26.0^\circ = 37.0^\circ$

so $|\vec{A}\times\vec{B}| = (5.00)(4.00)\sin 37.0^\circ = 12.04$

By the right hand-rule $\vec{A}\times\vec{B}$ is in the $+z$ direction, thus $(\vec{A}\times\vec{B})\cdot\vec{C} = (12.04)(6.00) = 72.2$